AE4115 Lab Exercise

Pre-Test Report

Course Code	AE4115	
Course Name	Experimental Simulations	
Student Name	Álvaro Rodríguez Santos	6420540
	Marcos Carrera Valle	5682525
	Max Comenge	5714672
	Nicolás García Arana	5461294
	Nicolás Nogueiras	5683408
Responsible Instructor	Dr.ir. T. Sinnige	
	Dr.ir. WJ Baars	
	Prof.Dr.-Ing. G. Eitelberg	
Measurement Challenge	MC1	
Report Date	January 20, 2026	

Part 1

This lab exercise aims to characterize the aerodynamic and acoustic behavior of a propeller-driven aircraft across a range of operational conditions. The primary objective is to quantify the influence of power effect on longitudinal stability, elevator effectiveness and trimmed performance. Additionally, the effects of boundary corrections, Reynolds number effects and noise emissions of the propeller will be studied.

1.1 Performance Indicators

The first step of the preparation is to define the performance indicators. Given the overall aim of quantifying power effects, the main power measurement is the advance ratio J which is a non dimensional parameter that describes the operating condition of the propeller. It is a function of velocity, rotational speed and propeller diameter. A high J demonstrates low thrust, a weak slipstream and approaches windmilling conditions.

To assess the longitudinal stability, the moment coefficient with respect to alpha C_{m_α}. The aircraft is considered stable as long as this value is below 0. It is found by measuring the longitudinal moment M at different angles of attack α.

Subsequently, the elevator effectiveness δ_e $C_{m_{\delta_e}}$ quantifies the authority of the elevator. Hence the longitudinal moment M with respect to the elevator deflection δ_e are required. Additionally, it is of interest to measure the control power, defined by the change in lift coefficient with respect to the elevator deflection $C_{L_{\delta_e}}$. For this, the lift force L shall be measured from the external balance, alongside δ_e.

To assess the aerodynamic performance in trimmed conditions the aerodynamic efficiency C_L/C_D is used as a performance indicator. Here, the lift and drag coefficient are determined from the lift L and drag force D respectively.

To quantify Reynolds effects on aerodynamic performance, the aforementioned forces and moments L, D and M are measured with changing α at several Reynolds numbers (hence several velocities V).

The aeroacoustic analysis employs physics-based non-dimensional scaling to characterize noise mechanisms while mitigating the impact of hard-walled test section reflections on absolute levels. The acquired voltage signals are calibrated into pressure fluctuations in the time-domain and transformed via FFT into frequency spectra, where the frequency is normalized to harmonic order (f/f_{BPF}) and the acoustic pressure (p_{rms}) is scaled by thrust and diameter into a non-dimensional Pressure Ratio ($\Pi_{noise} = p_{rms}D^2/T$). This approach facilitates the collapse of data from varying operating points onto a single curve.

The performance indicators are shown in table 1. Hence, we define L, D, M as the dependent variable which will be measured, and V, J, α, δ_e as the independent variables.

Table 1: Overview of Performance Indicators

Performance Indicator	**Symbol**	**Variables to measure**
Advance Ratio	J	V, n
Longitudinal Stability	C_{m_α}	M, α
Elevator Effectiveness	$C_{m_{\delta_e}}$	M, δ_e
Control Power	$C_{L_{\delta_e}}$	L, δ_e
Aerodynamic Efficiency	$\frac{C_L}{C_D}$	L, D
Aerodynamic Performance	$C_L(\alpha,\ Re), C_D(\alpha,\ Re)$, $C_M(\alpha,\ Re)$	$L,\ D,\ M, \alpha,\ V$
Noise	Π_{noise}	V_{noise}

1.2 Dimensional Analysis

To ensure that the data collected in the lab accurately predicts real flight performance, a systematic dimensional analysis is performed using the Buckingham π theorem.

The relevant parameters are identified in table 2, categorized by their experimental role. We then select three independent variables representing the mass M (ρ), time T (V), and length L (c) dimensions. By expressing the remaining parameters as dimensionless π-products and applying standard aerospace manipulations, we arrive at the coefficients used for analysis (tables 4 and 5).

Table 2: Experiment Parameters

Type	**Variable**	**Symbol (Units)**	**Dimensions**
Dependent Outputs	Lift Force	$L\ (N)$	$[MLT^{-2}]$
	Drag Force	$D\ (N)$	$[MLT^{-2}]$
	Pitching Moment	$M\ (N \cdot m)$	$[ML^2T^{-2}]$
	Pressure Fluctuations	$\Delta p\ (Pa)$	$[ML^{-1}T^{-2}]$
	Frequency of Microphone	$f\ (Hz)$	$[T^{-1}]$
Angles	Angle of Attack	$\alpha\ (rad)$	[1]
	Elevator Deflection	$\delta_e\ (rad)$	[1]
Air Properties	Velocity	$V\ (m/s)$	$[LT^{-1}]$
	Density	$\rho\ (kg/m^3)$	$[ML^{-3}]$
	Viscosity	$\mu\ (Pa \cdot s)$	$[ML^{-1}T^{-1}]$
	Speed of Sound	$a\ (m/s)$	$[LT^{-1}]$
	Reference Pressure	$p_{\text{ref}}\ (Pa)$	$[ML^{-1}T^{-2}]$
Prop. Force Output	Thrust	$T\ (N)$	$[MLT^{-2}]$
Prop. Configuration	Rotational Speed	$n\ (s^{-1})$	$[T^{-1}]$
Constants, Geometry	Mean Aerodynamic Chord	$c\ (m\ )$	$[L]$
Constants, Prop.	Wing Surface Area	$S\ (m^2)$	$[L^2]$
	Propeller Diameter	$D\ (m)$	$[L]$
	Number of Blades	$N\ (-)$	[1]

Table 3: Non-Dimensionality of Constants

Π-Product	Manipulation	Aerospace Convention	Name
$\Pi_I = D/c$	-	$GR = D/c$	Geometry Ratio
$\Pi_{II} = S/(c^2)$	$c = S/b$	$AR = b^2/S$	Aspect Ratio

Table 4: Non-dimensionalization of Variables

Π-Product	Manipulation	Aerospace Convention	Name
$\Pi_1 = \frac{L}{\rho V^2 c^2}$	$\Pi_1/(2\Pi_{II})$	$C_L = \frac{L}{\frac{1}{2}\rho V^2 S}$	Lift Coefficient
$\Pi_2 = \frac{D}{\rho V^2 c^2}$	$\Pi_2/(2\Pi_{II})$	$C_D = \frac{D}{\frac{1}{2}\rho V^2 S}$	Drag Coefficient
$\Pi_3 = \frac{M}{\rho V^2 c^3}$	$\Pi_3/(2\Pi_{II})$	$C_m = \frac{M}{\frac{1}{2}\rho V^2 S}$	Moment Coefficient
$\Pi_4 = \frac{T}{\rho V^2 c^2}$	$\Pi_4/(2\Pi_{II})$	$C_T = \frac{T}{\frac{1}{2}\rho V^2 S}$	Thrust coefficient
$\Pi_5 = \frac{n}{c^{-1}V}$	$1/(\Pi_5 \Pi_I)$	$J = \frac{V}{nD}$	Advance Ratio
$\Pi_6 = \frac{\mu}{\rho V c}$	$1/\Pi_6$	$Re_c = \frac{\rho V c}{\mu}$	Reynolds Number
$\Pi_7 = \frac{a}{V}$	$1/\Pi_7$	$Ma = \frac{V}{a}$	Mach Number

This dimensional analysis shows that the dependent aerodynamic outputs L, D and M can be represented as force and moment coefficients which are functions of C_T, J, Re_c, Ma, α and δ_e.

To perform the acoustic measurements, the acoustic dependent variables are also non-dimensionalized. The Blade Passage Frequency (BPF $= n \cdot N_B$) is introduced to aid in this process.

Table 5: Non-dimensionalization of Acoustic Variables

Π-Product	Manipulation	Aerospace Convention	Name
$\Pi_8 = \frac{p_{\text{ref}}}{V^2 \rho}$	$\sqrt{1/\Pi_8}$	$Eu = \frac{V}{\sqrt{p_{\text{ref}}/\rho}}$	Euler Number
$\Pi_9 = \frac{n}{c^{-1}V}$	-	$St = \frac{fL}{V}$	Strouhal Number
$\Pi_{10} = \frac{\Delta p}{V^2 \rho}$	$20 \cdot log_{10}(\Pi_{10}/\Pi_8)$	$\text{SPL} = 20 \cdot log_{10}(\frac{\Delta p}{p_{\text{ref}}})$	Sound Pressure Level
$\Pi_{11} = \frac{f}{c^{-1}V}$	$\Pi_{11}/(\Pi_9 N_b)$	$n_{\text{order}} = \frac{f}{\text{BPF}}$	Harmonic order

1.3 Scaling and Similarity

The wind-tunnel model is a geometrically scaled replica of the full-scale aircraft. To validate experimental data, the key scaling parameters must be addressed. The following scaling parameters have been identified in the dimensional analysis: Re, Ma, St. They are discussed below.

- **Reynolds Number** Re**:** In this experiment the Reynolds number will be much smaller than that of the real aircraft. In fact, a first estimate for the Reynolds number of the real aircraft and the scaled aircraft is $65.8 \cdot 10^6$ and $450 \cdot 10^6$ respectively - over 3 orders of magnitude in difference. This is because both the velocity (40 vs 240 m/s[1]) and the mean aerodynamic chord of the wing (0.165 vs 403 cm[2]) are

[1]Goldsmith, I. M., "A Study to Define the Research and Technology Requirements for Advanced Turbo/Propfan Transport Aircraft," NASA CR-166138, 1981.

[2]European Union Aviation Safety Agency , "Type-Certificate Data Sheet No. EASA IM.A.211 for MD-90-30, B717-200" tech. rep. (Issue 3.0) , EASA , 07 2025.

much smaller. To correct this, transition is forced on different surfaces using trip strips. Additionally, the Reynolds effects on aerodynamic performance are studied to extrapolate those trends to the full-scale condition.

- **Mach Number M:** Due to the facility's limitations, the maximum achievable Mach number is 0.12, far below the real Mach number of 0.8. Therefore in the test, compressibility effect are neglected, and only low-speed aerodynamic performance will be represented. This will evidently have an effect on the noise emissions analysis.

- **Strouhal Number St:** The Strouhal Number is the inverse of the advance ratio J, hence matching the advance ratio ensures Strouhal similarity, which is critical for accurate noise measurements.

1.4 Test setup and measurement objectives

The tests will be performed in the TU Delft Low-Turbulence Tunnel (LTT), a low-speed closed-return facility with a closed-wall test section (1.80 m $\times$ 1.25 m) and very low freestream turbulence ($< 0.05\%$). In the configuration used for this lab, the tunnel will be operated up to 40 m/s and is not acoustically treated, which is relevant for the aeroacoustic part of the measurements.
The aircraft model represents a regional configuration with an unswept wing and two tractor propellers mounted upstream of the horizontal tailplane. The model is mounted upside down and connected to an external 6-component balance through three support struts; angle of attack is adjusted via the aft strut, and sideslip is set by rotating the balance/turntable system. Transition is forced with trip strips on wing, tailplanes, fuselage and nacelles to promote turbulent-like behaviour at model scale.
Two main measurement systems are used:

1. External balance to obtain forces and moments (basis for stability/control derivatives and performance trends).

2. Inflow microphone (single location at $[X, Y, Z] = [55, 44, 43]$ cm relative to the starboard propeller) to measure acoustic pressure fluctuations in the form of voltage signals.

For measurement challenge 1 (longitudinal), the setup is intended to quantify: (i) power effects on longitudinal stability, (ii) elevator effectiveness/control power under slipstream influence, and (iii) performance in trimmed conditions, while also addressing boundary corrections, Reynolds-number effects, and propeller noise.

1.4.1 Key limitations and their impact on the results

1) Wind-tunnel wall interference (closed-wall)
The closed-wall test section induces blockage and streamline curvature, potentially biasing lift-curve slope, induced drag, and stability derivatives.
Mitigation: Apply boundary corrections in post-processing (comparing raw vs. corrected); use provided reference measurements ("tail-off" database) rather than reconfiguring the model during the slot.

2) Support-system interference (struts + inverted mount)

Struts add drag/wake effects, and the inverted mounting complicates interpretation. These add bias to moment measurements used for stability/control evaluation.
Mitigation: Rely on wind-off tares for every attitude to remove non-aerodynamic offsets. Avoid unnecessary attitude changes and use "check points" to detect drift.

3) Reynolds-number mismatch and forced transition
Full-scale Reynolds similarity is unreachable. Forced transition (trip strips) improves representativeness.
Mitigation: (i) Focus analysis on non-dimensional trends (power-on vs. power-off deltas).

4) Restricted envelope and discrete settings
Constraints ($V_\infty \leq 40$ m/s, $-5° \leq \alpha \leq +14°$, $-10° \leq \beta \leq +10°$, discrete elevator and rudder settings, and a fixed propeller blade pitch) reduce longitudinal trim space coverage and derivative mapping resolution.

5) Aeroacoustic limitations (hard-wall reflections)
Reflections in the untreated test section affect absolute noise levels and mask broadband content.
Mitigation: This is imposed; no solution or reduction of effects is possible.

1.4.2 Time constraints during execution

In addition to the aerodynamic and measurement limitations discussed above, the execution of the test is strongly constrained by the available tunnel time and by the downtime required for configuration changes. These constraints directly affect the achievable number of operating points, the extent of high-lift exploration, and the feasibility of dedicated "component-off" datasets within the allocated slot.

- **Initial preparation time:** Approximately 10 minutes are reserved at the start of the session for preparation and familiarization, reducing the effective time available for measurements from 1 h 45 min to ∼1 h 35 min.

- **Tunnel start-up overhead:** Starting (or restarting) the tunnel typically requires 3–5 minutes. This overhead becomes especially important if the test is interrupted by configuration changes.

- **Freestream condition changes:** Adjusting the freestream velocity typically takes 1–2 minutes. Although it may reduce to ∼30 seconds as the session progresses, a minimum of ∼1 minute should be assumed.

- **Attitude changes (α, β):** Once a new condition is commanded, the mechanical/operational response time is typically 10–20 seconds, with additional rate limits of approximately 0.5°/s for angle of attack and 0.25°/s for sideslip. This favours test matrices built around smooth sweeps (small increments) rather than large jumps.

- **Elevator configuration changes:** Changing the elevator setting on the model requires ∼10 minutes and effectively forces the test to stop (wind-off) and later restart. This makes frequent elevator changes impractical and strongly motivates using a small number of elevator settings.

- **Rudder configuration changes:** Changing the rudder configuration similarly requires ∼10 minutes and forces a stop/restart. Rudder changes should therefore be minimized and, when necessary, grouped into one dedicated block of testing.

Overall mitigation strategy: given these time constraints, the test matrix should be designed to (i) minimize wind-off configuration changes, (ii) group points by freestream speed and propeller setting to reduce stabilization overhead, (iii) use smooth attitude sweeps rather than scattered points, and (iv) rely on the provided "component-off" datasets (propeller-off and HTP-off) for baseline comparisons, focusing tunnel time on the unique power-on and control-effectiveness measurements needed to meet the objectives.

1.4.3 Propeller thrust estimation from external balance data

Propeller thrust will be estimated indirectly from the external balance by comparing power-on measurements with the provided propeller-off reference data. For each test point, an equivalent propeller-off axial force X_{off} will be obtained by interpolating the "propOff" dataset to the same $(V_\infty, \alpha, \beta, \delta)$. The effective thrust is then inferred from the axial-force increment between power-on and propeller-off conditions, i.e.

$$T_{\text{est}} = -\left(X_{\text{on}} - X_{\text{off}}\right),$$

or equivalently using force coefficients as $T_{\text{est}} = -\Delta C_X q_\infty S$, with the sign convention consistent with the adopted wind-axis system.
This approach provides the net propulsive contribution in the axial force balance, while differences observed in normal force and pitching moment between power-on and propeller-off conditions are attributed to slipstream–airframe interaction and treated separately in the aerodynamic analysis.

Part 2

Because the experiment is time-bounded, the productivity and quality of data acquisition must be optimized rather than the conventional focus on quantity without direction. Because of this, the test has to be planned before hand. A test matrix ensures all measurements required for this assignment are captured whilst minimizing downtime and systematic errors. The reasoning behind the test matrix is presented in this part, and the full test matrix is presented in appendix A as figs. 7, 8 and 9.

2.1 Selection of Test Points

It is clear from the previous part that the dependent variables to measure are C_m, C_L, C_D, SPL and n_{order} for a range of independent variables Re, J, α and δ_e. Note that to change the Reynolds number and advance ratio one must change the velocity V and the propeller rotational speed n respectively.

We can optimize how we change the independent variables to reduce the downtime. Fig. 6 shows how long each variable takes to change. From this table, the following hierarchy can be set: δ_e, Re (V), J (n), α. Hence, it is decided to create blocks for δ_e, V and J. Within each δ_e-block, V is changed. Withing each V-block, J changes, and within each J-block, α changes. The order in which these variables change within each block is discussed in section 2.2.

The next step is to decide on how many data points per independent variable to take. We are of course constrained by the time limit. We let n_δ, n_V, n_J and n_α be the number of points measure for each variable. A minimum of 3 points for δ_e and α are required to identify any non-nonlinearities in the graphs for the lift, drag and moment polars, longitudinal stability (C_{m_α}), elevator effectiveness ($C_{m_{\delta_e}}$) and elevator control power ($C_{L_{\delta_e}}$) analyses. n_J defines the number of points in our $C_{m_\alpha} - J$, $C_{m_{\delta_e}} - J$, $C_{L_{\delta_e}} - J$ and $\delta_{\text{e, trim}} - J$ graphs, whilst n_V defines the number of curves in each graph. It is deemed that the number of points is more important than the number of graphs. With trial and error, the following optimal number of data points is chosen:

Table 6: Time to Change and Number of Data Points per Variable

Independent Variable, i	$\Delta t_{i,\,\textbf{approx}}$ (min)	n_i
δ_e	10	3
V	1	2
J	0.75	3
α	0.55	3

To determine which values are used for each variable, we analyze each individually:

- **Elevator Deflection** δ_e (deg)**:** $(-5, 0, 5)$
  $0°$ serves as the baseline condition and is a requirement for the acoustic evaluation (Manual, pg. 11[3]). The elevator deflection angle doesn't vary signifcantly during typicall operation, hence the $\pm 5°$ range is representative.

- **Velocity** V (m/s)**:** (20, 40)
  The maximum velocity of 40 m/s is chosen to reach the highest possible Reynolds number (Re) within the constraints, and hence maximizing similarity with the full-scale model. 20 m/s has imposed as the limit minimum velocity of the tunnel. This corresponds to a Reynolds number of $2.13 \cdot 10^5$ to $4.26 \cdot 10^5$.

- **Advance Ratio** J $(-)$**:** (1.6, 1.9, 2.2)
  $J = 1.6$ is required for the acoustic evaluation and is set as the lower bound for J. To understand how the propeller performance scales, we examined the provided C_T data (Manual, fig. 11), which shows valid results up to $J \approx 2.2$ for $V_\infty =$ 20 m/s. Hence this is selected as the upper bound. The final value is selected at the midpoint.

- **Angle of Attack** α (deg)**:** $(-4, 0, 8)$
  An angle of attack of $0°$ represents the typical cruise condition. $8°$ represents a climb condition within the linear regime, and is required for the acoustic evaluation. Finally, it was decided that covering the negative angles of attack was more important than having an even larger α, as the typical range of α for a turboprop regional jet doesn't increase beyond $10°$, and longitudinal stability is very important during high speed descents where the angle of attack may become negative.

The selection of these points also ensures that the mandatory test condition for acoustic evaluation ($V = 40$ [m/s], $J = 1.6$ [$-$], $\alpha = 8$ [°], $\delta_e = 0$ [°]) is ran.

[3] T. Sinnige, W. J. Baars, and G. Eitelberg, "Lab-Exercise Manual - AE4115 Experimental Simulations," tech. rep., TU Delft, Delft, 11 2025

2.2 Error Mitigation

Experimental measurements are susceptible to both random errors and systematic errors. To ensure the quality of the results, quality assurance techniques are used to counteract these different error types.

Firstly, blocking can be used to partially avoid the impact of bias shifts over time in experimental data. Within our experiment, we have already identified three blocks: δ_e, V and J-blocks. Now, because changing the elevator deflection angle is extremely time consuming, blocking cannot be applied. However, this is not the case for the V and J-blocks. Using a python script, it has been seen that the order of the test points can be changed with only a X min time penalty, which allows us to remain within the given time. Therefore, instead of running the blocks in increasing V or J, we ensure the order is "randomized".

Replication is applied by returning to the original conditions at the end of each δ_e-block (points 1*, 19*, and 37*). Replication usually entails repeating a measurement under identical conditions to quantify random error and improve the precision of the mean result. However due to time limits, only a total of 3 measurements will be "repeated", and the function of this is to check for instrumental/ environmental drift between the start and end of δ_e-blocks rather than improving the precision of the mean result.

Finally, randomization is also utilized to convert potentially biased systematic errors into random noise. To apply this, the angles of attack and advance ratios are selected in a random order rather than following a monotonic sequence. Although the design freedom is limited because there are only three points per variable, randomization can still be achieved by ensuring that the specific sequence of values is not repeated identically across different J-blocks and V-blocks. This prevents time-dependent patterns from being mistakenly interpreted as aerodynamic effects.

2.3 Contingency Plans

A contingency plan is required in case the test matrix does not correctly estimate the time required for measurements. Two different scenario's become possible; either the measurements are carried out quicker than expected or they take longer than expected. In the former, which is more unlikely, more measurements can be carried out. To attain a more accurate lift polar, higher angles of attack will be measured at an elevator deflection of 0 (note the last block of the matrix will correspond to this angle in order to evade wasting time in this scenario). Also, the maximum velocity will be used so that the conditions assimilate the flight condition at which the aircraft would fly. The potential extension of the test matrix is seen after the test matrix in Table 10. In the second scenario, which is more likely as human errors generally occur in experimental procedures, the contingency plan comes from the matrix layout. There is already a 10 minute advantage which the measurements can be delayed by and still be within the time constraints. However, at the end of the first block, a measurement of the used time will be compared to the expected time. In case the measured time heavily exceeds the expected time, no measurements will be repeated from the second block forward.

A Full Test Matrix

Table 7: Full Test Matrix ($\delta_e = 0$)

Point #	**Action**	$\boldsymbol{\delta_e}$ **(deg)**	V	J $(-)$	α (deg)	ΔT (min)	T_{tot} (min)
-	Introduction					10	10
-	Start Test	0	20	1.6	0	2	12
1	Measure	0	20	1.6	0	0.50	12.50
-	Change a	0	20	1.6	8	0.517	13.017
2	Measure	0	20	1.6	8	0.50	13.517
-	Change a	0	20	1.6	-4	0.65	14.167
3	Measure	0	20	1.6	-4	0.50	14.667
-	Change J	0	20	1.9	-4	0.75	15.417
4	Measure	0	20	1.9	-4	0.50	15.917
-	Change a	0	20	1.9	8	0.65	16.567
5	Measure	0	20	1.9	8	0.50	17.067
-	Change a	0	20	1.9	0	0.517	17.583
6	Measure	0	20	1.9	0	0.50	18.083
-	Change J	0	20	2.2	0	0.75	18.833
7	Measure	0	20	2.2	0	0.50	19.333
-	Change a	0	20	2.2	-4	0.383	19.717
8	Measure	0	20	2.2	-4	0.50	20.217
-	Change a	0	20	2.2	8	0.65	20.867
9	Measure	0	20	2.2	8	0.50	21.367
-	Change V	0	40	2.2	8	1.00	22.367
10	Measure	0	40	2.2	8	0.50	22.867
-	Change a	0	40	2.2	0	0.517	23.383
11	Measure	0	40	2.2	0	0.50	23.883
-	Change a	0	40	2.2	-4	0.383	24.267
12	Measure	0	40	2.2	-4	0.50	24.767
-	Change J	0	40	1.9	-4	0.75	25.517
13	Measure	0	40	1.9	-4	0.50	26.017
-	Change a	0	40	1.9	0	0.383	26.400
14	Measure	0	40	1.9	0	0.50	26.900
-	Change a	0	40	1.9	8	0.517	27.417
15	Measure	0	40	1.9	8	0.50	27.917
-	Change J	0	40	1.6	8	0.75	28.667
16	Measure	0	40	1.6	8	0.50	29.167
-	Change a	0	40	1.6	-4	0.65	29.817
17	Measure	0	40	1.6	-4	0.50	30.317
-	Change a	0	40	1.6	0	0.383	30.700
18	Measure	0	40	1.6	0	0.50	31.200
-	Change V	0	20	1.6	0	1.00	32.200
1*	Measure	0	20	1.6	0	0.50	32.700

Table 8: Full Test Matrix ($\delta_e = 5$)

Point #	**Action**	$\boldsymbol{\delta_e}$ **(deg)**	V	J $(-)$	α (deg)	ΔT (min)	T_{tot} (min)
-	Change e	5	40	1.6	0	10.000	42.700
19	Measure	5	40	1.6	0	0.500	43.200
-	Change a	5	40	1.6	8	0.517	43.717
20	Measure	5	40	1.6	8	0.500	44.217
-	Change a	5	40	1.6	-4	0.650	44.867
21	Measure	5	40	1.6	-4	0.500	45.367
-	Change J	5	40	2.2	-4	0.750	46.117
22	Measure	5	40	2.2	-4	0.500	46.617
-	Change a	5	40	2.2	8	0.650	47.267
23	Measure	5	40	2.2	8	0.500	47.767
-	Change a	5	40	2.2	0	0.517	48.283
24	Measure	5	40	2.2	0	0.500	48.783
-	Change J	5	40	1.9	0	0.750	49.533
25	Measure	5	40	1.9	0	0.500	50.033
-	Change a	5	40	1.9	-4	0.383	50.417
26	Measure	5	40	1.9	-4	0.500	50.917
-	Change a	5	40	1.9	8	0.650	51.567
27	Measure	5	40	1.9	8	0.500	52.067
-	Change V	5	20	1.9	8	1.000	53.067
28	Measure	5	20	1.9	8	0.500	53.567
-	Change a	5	20	1.9	0	0.517	54.083
29	Measure	5	20	1.9	0	0.500	54.583
-	Change a	5	20	1.9	-4	0.383	54.967
30	Measure	5	20	1.9	-4	0.500	55.467
-	Change J	5	20	2.2	-4	0.750	56.217
31	Measure	5	20	2.2	-4	0.500	56.717
-	Change a	5	20	2.2	0	0.380	57.100
32	Measure	5	20	2.2	0	0.500	57.600
-	Change a	5	20	2.2	8	0.520	58.117
33	Measure	5	20	2.2	8	0.500	58.617
-	Change J	5	20	1.6	8	0.750	59.367
34	Measure	5	20	1.6	8	0.500	59.867
-	Change a	5	20	1.6	-4	0.650	60.517
35	Measure	5	20	1.6	-4	0.500	61.017
-	Change a	5	20	1.6	0	0.380	61.400
36	Measure	5	20	1.6	0	0.500	61.900
-	Change V	5	40	1.6	0	1.000	62.900
19*	Measure	5	40	1.6	0	0.500	63.400

Table 9: Full Test Matrix ($\delta_e = -5$)

Point #	**Action**	$\boldsymbol{\delta_e}$ **(deg)**	V	J $(-)$	α (deg)	ΔT (min)	T_{tot} (min)
-	Change e	-5	20	1.6	0	10.000	73.400
37	Measure	-5	20	1.6	0	0.500	73.900
-	Change a	-5	20	1.6	-4	0.383	74.283
38	Measure	-5	20	1.6	-4	0.500	74.783
-	Change a	-5	20	1.6	8	0.650	75.433
39	Measure	-5	20	1.6	8	0.500	75.933
-	Change J	-5	20	1.9	8	0.750	76.683
40	Measure	-5	20	1.9	8	0.500	77.183
-	Change a	-5	20	1.9	0	0.517	77.700
41	Measure	-5	20	1.9	0	0.500	78.200
-	Change a	-5	20	1.9	-4	0.383	78.583
42	Measure	-5	20	1.9	-4	0.500	79.083
-	Change J	-5	20	2.2	-4	0.750	79.833
43	Measure	-5	20	2.2	-4	0.500	80.333
-	Change a	-5	20	2.2	0	0.383	80.717
44	Measure	-5	20	2.2	0	0.500	81.217
-	Change a	-5	20	2.2	8	0.517	81.733
45	Measure	-5	20	2.2	8	0.500	82.233
-	Change V	-5	40	2.2	8	1.000	83.233
46	Measure	-5	40	2.2	8	0.500	83.733
-	Change a	-5	40	2.2	-4	0.650	84.383
47	Measure	-5	40	2.2	-4	0.500	84.883
-	Change a	-5	40	2.2	0	0.383	85.267
48	Measure	-5	40	2.2	0	0.500	85.767
-	Change J	-5	40	1.9	0	0.750	86.517
49	Measure	-5	40	1.9	0	0.500	87.017
-	Change a	-5	40	1.9	8	0.520	87.533
50	Measure	-5	40	1.9	8	0.500	88.033
-	Change a	-5	40	1.9	-4	0.650	88.683
51	Measure	-5	40	1.9	-4	0.500	89.183
-	Change J	-5	40	1.6	-4	0.750	89.933
52	Measure	-5	40	1.6	-4	0.500	90.433
-	Change a	-5	40	1.6	8	0.650	91.083
53	Measure	-5	40	1.6	8	0.500	91.583
-	Change a	-5	40	1.6	0	0.520	92.100
54	Measure	-5	40	1.6	0	0.500	92.600
-	Change V	-5	20	1.6	0	1.000	93.600
37*	Measure	-5	20	1.6	0	0.500	94.100

Contingency Test Matrix Extension in case of more time:

Table 10: Full Test Matrix ($\delta_e = 0$)

Point #	**Action**	$\boldsymbol{\delta_e}$ **(deg)**	V	J $(-)$	α (deg)	ΔT (min)	T_{tot} (min)
-	Change a	0	20	1.6	10	0.583	94.683
-	Change V	0	40	1.6	10	1.000	95.683
-	Measure	0	40	1.6	10	0.500	96.183
-	Change a	0	40	1.6	15	0.417	96.600
-	Measure	0	40	1.6	15	0.500	97.100
-	Change a	0	40	1.6	12	0.350	97.450
-	Measure	0	40	1.6	12	0.500	97.950
-	Change a	0	40	1.6	14	0.317	98.267
-	Measure	0	40	1.6	14	0.500	98.767
-	Change a	0	40	1.6	13	0.283	99.050
-	Measure	0	40	1.6	13	0.500	99.550
-	Change J	0	40	2.2	13	0.750	100.300
-	Measure	0	40	2.2	13	0.500	100.800
-	Change a	0	40	2.2	12	0.283	101.083
-	Measure	0	40	2.2	12	0.500	101.583
-	Change a	0	40	2.2	14	0.317	101.900
-	Measure	0	40	2.2	14	0.500	102.400
-	Change a	0	40	2.2	10	0.383	102.783
-	Measure	0	40	2.2	10	0.500	103.283
-	Change a	0	40	2.2	15	0.417	103.700
-	Measure	0	40	2.2	15	0.500	104.200

B Statement of Fair Conduct

We hereby declare that:

- I have fairly contributed, in the measure of at least $\approx 20\%$ to the generation of all the parts of the delivered material,
- All the delivered material is the result of the work performed by my team and no others,
- I am aware of the consequences of committing fraud[4] (i.e. piggybacking, copying, fiddling with data and plagiarism).

Group Number: 8

Student 1		**Student 2**	
Given name	Álvaro	Given name	Marcos
Family name	Rodríguez Santos	Family name	Carrera Valle
Student number	6420540	Student number	5682525
Date	January 20, 2026	Date	January 20, 2026
Place	Delft	Place	Delft
Signature		Signature	

Student 3		**Student 4**	
Given name	Maximiliano	Given name	Nicolás
Family name	Comenge	Family name	García Arana
Student number	5714672	Student number	5461294
Date	January 20, 2026	Date	January 20, 2026
Place	Delft	Place	Delft
Signature		Signature	

Student 5	
Given name	Nicolás
Family name	Nogueiras Garcia
Student number	5683408
Date	January 20, 2026
Place	Delft
Signature	

[4] `https://www.tudelft.nl/en/student/my-study-me/rules-guidelines-and-participation/fraud-plagiarism`